



ΣΥΣΤΗΜΑΤΑ ΔΙΑΧΕΙΡΙΣΗΣ ΚΑΙ ΑΝΑΛΥΣΗΣ ΔΕΔΟΜΕΝΩΝ

ΕΑΡΙΝΟ ΕΞΑΜΗΝΟ 2023-2024

ΓΙΩΡΓΟΣ ΚΟΥΡΟΣ

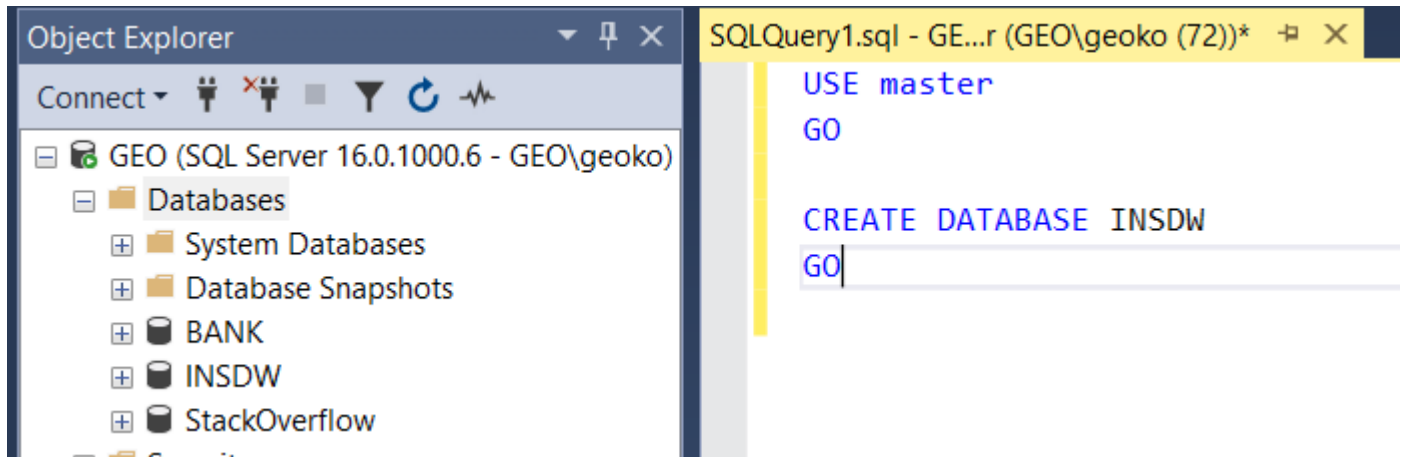
ΑΜ : 3190095

ΔΕΥΤΕΡΗ ΕΡΓΑΣΙΑ



Ζήτημα Πρώτο

1. Δημιουργία βάσης INSDW, πίνακα inspections_data και εισαγωγή των δεδομένων από το inspections_data.txt αρχείο.



```
CREATE TABLE inspections_data (
    rid int,
    lat float,
    lon float,
    insdate date,
    insyear int,
    insmonth int,
    insday int,
    insweekday int,
    inscode int,
    instype nvarchar(100),
    criticalissue int,
    nonCriticalissue int,
    vcode int,
    vdescription nvarchar(255),
    vcategory nvarchar(255)
)
```

```
bulk insert inspections_data
from 'C:\Users\geoko\Desktop\inspections_data\inspections_data.txt'
with (datafiletype = 'widechar', firstrow = 2, fieldterminator = '|',
rowterminator = '\n');
```

2. Υλοποίηση λογικού σχήματος της αποθήκης δεδομένων μορφής αστέρα (Star schema) με fact table τον πίνακα ins_data και 4 dimension πίνακες (violations, inspection_type, timeinfo, restaurants)

```
create table violations (
    vcode int primary key,
    vdescription nvarchar(255),
    vcategory nvarchar(255)
)

create table inspection_type (
    inscode int primary key,
    instype nvarchar(100)
)

create table timeinfo (
    insdate date primary key,
    insyear int,
    insmonth int,
    insday int,
    insweekday int
)

create table restaurants (
    rid int primary key,
    lat float,
    lon float
)

create table ins_data (
    rid int,
    insdate date,
    inscode int,
    vcode int,
    criticalissue int,
    nonCriticalissue int,

    primary key (rid, insdate, inscode, vcode),
    foreign key (rid) references restaurants(rid),
    foreign key (insdate) references timeinfo(insdate),
    foreign key (inscode) references inspection_type(inscode),
    foreign key (vcode) references violations(vcode)
)
```

3. Εντολές SQL για τροφοδότηση δεδομένων του παραπάνω σχήματος από τον πίνακα inspections_data

```
insert into violations select distinct vcode, vdescription, vcategory from inspections_data order by vcode

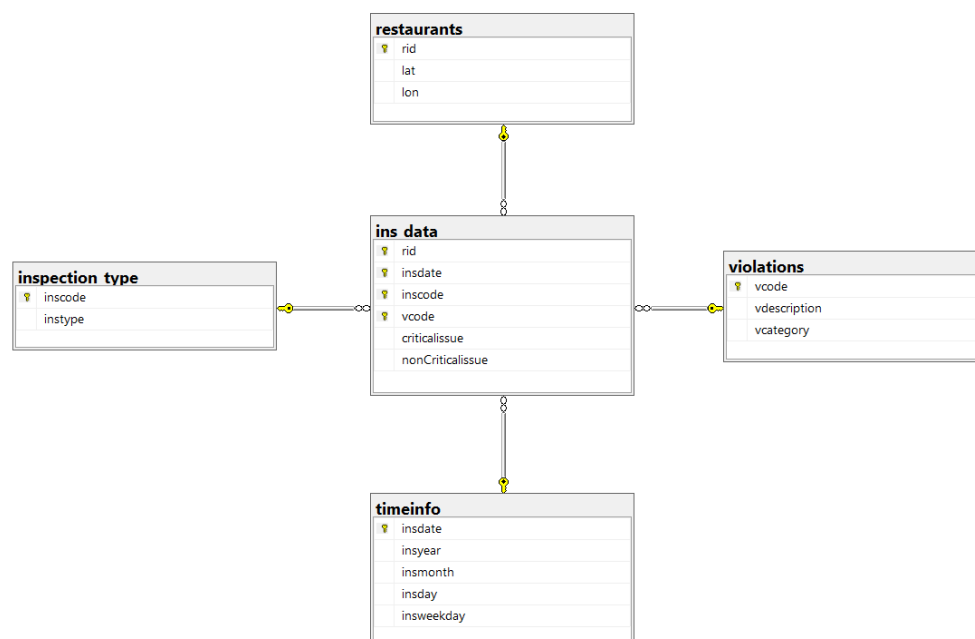
insert into timeinfo select distinct insdate, insyear, insmonth, insday, insweekday from inspections_data order by insdate

insert into restaurants select distinct rid, lat, lon from inspections_data order by rid

insert into inspection_type select distinct inscode, instype from inspections_data order by inscode

insert into ins_data select rid, insdate, inscode, vcode, criticalissue, nonCriticalissue from inspections_data
```

- 4.



Ζήτημα Δεύτερο

1.

```
select timeinfo.insyear, inspection_type.instype, count(*) as num_of_inspections
from ins_data, timeinfo, inspection_type
where ins_data.insdate = timeinfo.insdate and inspection_type.inscode = ins_data.inscode
group by rollup(timeinfo.insyear, inspection_type.instype)
order by timeinfo.insyear desc;
```

2.

```
select restaurants.rid, lat, lon, sum(ins_data.nonCriticalissue) as NonCriticalIssues,
    sum(ins_data.criticalissue) as CriticalIssues,
    (sum(ins_data.criticalissue) + sum(ins_data.nonCriticalissue)) as TotalIssues
from restaurants, ins_data where ins_data.rid = restaurants.rid
group by restaurants.rid, lat, lon
order by TotalIssues desc;
```

3.

```
select timeinfo.insyear, inspection_type.instype, violations.vcategory, sum(ins_data.criticalissue) as TotalCriticalIssues
from timeinfo, inspection_type, violations, ins_data
where timeinfo.insdate = ins_data.insdate and inspection_type.inscode = ins_data.inscode
    and violations.vcode = ins_data.vcode
group by cube(timeinfo.insyear, inspection_type.instype, violations.vcategory)
order by timeinfo.insyear desc, inspection_type.instype, violations.vcategory
```

Χρησιμοποιήθηκε order by για εκτενέστερη κατανόηση και debug των αποτελεσμάτων (δεν είναι απαραίτητο).